THE ICON ACADEMY
12th Class Computer Science
Exercise Short Questions & Answers

THE ICON ACADEMY

CHAPTER 1 : DATA BASICS

1. Difference between Data and Information with the help of example?
Ans: Data consists of raw facts and figures that have no meaning by themselves. Information is processed data that becomes meaningful and useful. For example, students' marks are data, while the final result sheet is information.

2. Describe Data Capturing?
Ans: Data capturing is the process of collecting and entering data into a computer system. It can be done manually through keyboard entry or automatically using devices such as scanners and barcode readers.

3. Difference between DDL and DML?
Ans: DDL (Data Definition Language) is used to define and modify database structure, such as creating or deleting tables. DML (Data Manipulation Language) is used to insert, update, delete, and retrieve data from tables.

4. Describe Data Manipulation?
Ans: Data manipulation means performing operations on data stored in a database. It includes adding new records, updating existing records, deleting records, and retrieving required information.

5. What is File? OR Define Data Set?
Ans: A file is a collection of related records stored together under a single name. It is used to organize and manage data for easy access and processing.

6. List the File Types from Usage Point of View?
Ans: From the usage point of view, files are mainly of two types: Program Files and Data Files. Program files contain instructions, while data files store actual information.

7. Difference between Program File and Data File?
Ans: A program file contains a set of instructions written in a programming language. A data file stores facts and figures that are processed by the program.

8. List any Two Problems in Traditional File Approach?
Ans: The traditional file approach causes data redundancy, where the same data is stored multiple times. It also creates data inconsistency, leading to conflicting information.

9. Difference between Master File and Transaction File?
Ans: A master file contains permanent data, such as employee or customer records. A transaction file stores temporary data related to daily business activities.

10. Difference between Direct Access and Sequential File?

Ans: In direct access, records can be accessed immediately without reading previous records. In sequential access, records are accessed one after another in a fixed order.

11. What is Meant by File Organization?

Ans: File organization refers to the method used to arrange records in a file. It determines how data is stored, retrieved, and updated efficiently.

12. What is the Main Purpose of Database System?

Ans: The main purpose of a database system is to store, organize, and manage data efficiently. It allows quick retrieval, updating, and sharing of information.

13. Difference between Database and Database System with Example?

Ans: A database is an organized collection of related data, such as a student record database. A database system includes the database, DBMS, hardware, software, and users.

14. Difference between Data Redundancy and Data Consistency?

Ans: Data redundancy means storing the same data in multiple places unnecessarily. Data consistency ensures that the same data remains accurate and identical throughout the database.

15. Define Data Integrity.

Ans: Data integrity refers to the accuracy, completeness, and reliability of data in a database. It ensures that data remains correct and unchanged during operations.

16. Define Hierarchical Model.

Ans: The hierarchical model organizes data in a tree structure where each child has only one parent. The network model allows a child to have multiple parents, creating a more flexible structure.

17. Define DBMS and Write any Two Objectives of DBMS?

Ans: DBMS stands for Database Management System, which is software used to manage databases. Its objectives are to reduce data redundancy and provide secure access to data.

18. Write any Two Advantages of DBMS?

Ans: DBMS improves data security by controlling user access. It also reduces data duplication and ensures better data consistency.

19. Define Data Dictionary and Write any Two Uses?

Ans: A data dictionary is a repository that stores information about database objects such as tables, fields, and relationships. It helps in database design and documentation.

20. List any Two Features of DBMS?

Ans: DBMS provides data security and supports data sharing among multiple users. It also offers backup and recovery facilities.

21. Why is Report Generator Used in Database System?

Ans: A report generator is used to create formatted reports from database data. It helps users analyze, print, and present information in a structured manner.

CHAPTER 4 : DATA INTEGRITY & NORMALIZATION

1. Define Database Integrity?

Ans: Database integrity refers to the accuracy, consistency, and reliability of data stored in a database. It ensures that data remains correct and valid throughout its lifecycle.

2. Define Normalization?

Ans: Normalization is the process of organizing data in a database to reduce redundancy and improve data consistency. It divides large tables into smaller related tables.

3. How Database Integrity is Maintained?

Ans: Database integrity is maintained by using constraints such as primary key, foreign key, and unique constraints. It also ensures proper validation and rules during data entry.

4. How First and Second Normal Form Achieved?

Ans: First Normal Form removes repeating groups and ensures atomic values. Second Normal Form removes partial dependency by ensuring non-key attributes depend on the whole primary key.

5. Define Referential Integrity. When it Used?

Ans: Referential integrity ensures that relationships between tables remain consistent. It is used when foreign keys in one table correctly match primary keys in another.

6. Difference between Entity Integrity and Data Integrity?

Ans: Entity integrity ensures that each table has a unique primary key with no null values. Data integrity ensures overall accuracy and consistency of all data in the database.

7. How do we Attain Entity Integrity in Database?

Ans: Entity integrity is achieved by defining a primary key in each table. The primary key must have unique and non-null values for every record.

8. How Referential Integrity can be Achieved?

Ans: Referential integrity is achieved by using foreign keys that match primary keys in related tables. It prevents invalid or orphan records in the database.

9. Define Functional Dependency with Example?

Ans: Functional dependency occurs when one attribute uniquely determines another attribute. For example, Student ID determines Student Name in a student table.

10. Difference between Full Functional and Transitive Dependency?

Ans: Full functional dependency occurs when a non-key attribute depends on the whole primary key. Transitive dependency occurs when a non-key attribute depends on another non-key attribute.

11. Difference between Synonym and Homonym?

Ans: A synonym means different names used for the same data item. A homonym means the same name is used for different data items.

12. Define Partial Dependency?

Ans: Partial dependency occurs when a non-key attribute depends on part of a composite primary key. It violates second normal form.

13. What are the Database Anomalies? List their Name?

Ans: Database anomalies are problems that occur in poorly designed databases. They include insertion anomaly, deletion anomaly, and update anomaly.

14. Define Repeating Group?

Ans: A repeating group is a set of multiple values stored in a single field. It violates the rules of first normal form.

15. Difference between 3rd and 4th Normal Form?

Ans: Third Normal Form removes transitive dependencies from a table. Fourth Normal Form removes multi-valued dependencies.

16. Define Transitive Dependency?

Ans: Transitive dependency occurs when a non-key attribute depends on another non-key attribute. It leads to redundancy and inconsistency.

17. Define Mutual Exclusiveness of Data?

Ans: Mutual exclusiveness means that two data values cannot exist at the same time. If one value is selected, the other must be excluded.

18. Define the Reproduction in Database?

Ans: Reproduction in database means duplication of data in multiple places. It increases redundancy and may cause inconsistency if not controlled.

CHAPTER 8 : GETTING STARTED WITH "C"

1. What is computer program?

Ans: A computer program is a set of instructions written in a programming language to perform a specific task. It tells the computer how to process data and produce results.

2. What is programming language?

Ans: A programming language is a formal language used to write instructions for a computer. It allows humans to communicate with machines.

3. Difference between High level language and low level language with example?

Ans: High-level language is easy to read and understand (e.g., C, Java). Low-level language is closer to machine code (e.g., Assembly language).

4. Define the term IDE with example?

Ans: IDE (Integrated Development Environment) is a software tool used to write, compile, and run programs. Example: Turbo C++ and Visual Studio.

5. Difference between Source Code and Object Code?

Ans: Source code is the original program written by the programmer. Object code is the machine-readable output produced after compilation.

6. Difference between Compiler and Interpreter?

Ans: A compiler translates the whole program at once into machine code. An interpreter translates and executes code line by line.

7. Write any three characteristics of C language?

Ans: C language is simple, fast, and portable. It also supports structured programming and efficient memory management.

8. Write the use of Turbo C++?

Ans: Turbo C++ is used to write, compile, and execute C and C++ programs. It provides an easy interface for beginners.

9. Difference between Linking and Loading?

Ans: Linking combines object files into a single executable program. Loading transfers the program into memory for execution.

10. Difference between Linker and Loader?

Ans: A linker combines different program modules. A loader loads the executable program into memory for execution.

11. What is meant by structured programming language?

Ans: Structured programming language organizes code into functions and blocks. It improves readability and reduces complexity.

12. What is purpose of include directive?

Ans: The include directive is used to add header files in a program. It allows reuse of predefined functions.

13. What is #define directive? With syntax and example?

Ans: It is used to define constants or macros in C programming. Syntax: #define PI 3.14

14. What is preprocessor directive?

Ans: Preprocessor directives are commands processed before compilation. They start with # symbol, like #include and #define.

15. What is header file?

Ans: A header file contains predefined functions and declarations. It is included using #include directive.

16. Define delimiter?

Ans: A delimiter is a symbol used to separate elements in a program. Example: comma, semicolon, or space.

17. What is statement terminator?

Ans: A statement terminator is a symbol used to end a statement in C language. In C, semicolon (;) is used.

18. Difference between Bug and Debug in C?

Ans: A bug is an error in a program. Debugging is the process of finding and fixing those errors.

19. Difference between Machine Language and Assembly Language?

Ans: Machine language uses binary code (0s and 1s). Assembly language uses symbolic instructions like ADD, MOV.

20. Difference between Syntax and Logical Errors?

Ans: Syntax errors occur due to incorrect grammar in code. Logical errors produce incorrect output despite correct syntax.

21. Why does ML Program execute faster? Give reason?

Ans: Machine language programs execute faster because they are directly understood by the computer CPU without translation.

22. Why C is a strongly typed language?

Ans: C is strongly typed because every variable must have a defined data type. It ensures type safety during execution.

23. Difference between Structured and Unstructured Programming?

Ans: Structured programming uses functions and blocks for organization. Unstructured programming uses random flow without proper structure.

CHAPTER 10 : INPUT / OUTPUT

1. Difference between standard input and output?

Ans: Standard input is used to take data from the user. Standard output is used to display results on the screen.

2. List some important input and output function in C?

Ans: Important functions are printf(), scanf(), getch(), and puts(). They handle input and output operations.

3. Difference between \n and \t escape sequence?

Ans: \n is used for new line. \t is used for tab space.

4. Difference between format specifiers and field width specifier?

Ans: Format specifiers define data type like %d, %f. Field width specifier controls spacing and alignment of output.

5. printf("\n \n") write output?

Ans: * (new line) ** — The \n moves to a new line before each * so stars appear on separate lines with an extra line in between.

6. List four types of format specifier?

Ans: Common format specifiers are %d, %f, %c, %s.

7. Difference between printf() and scanf() in C?

Ans: printf() is used for output. scanf() is used for input.

8. Define escape sequence. Also write its purpose?

Ans: Escape sequences are special characters starting with \. They are used for formatting output like new line or tab.

9. Why ampersand (&) used in scanf()?

Ans: It is used to provide memory address of variable. It allows input value to be stored correctly.

10. Which type of error is difficult to trace and why?

Ans: Logical errors are difficult to trace because program runs but gives incorrect output.

11. Difference between getch() and getche()?

Ans: getch() does not display input character. getche() displays the character on screen.

12. Define clrscr() function in C?

Ans: clrscr() is used to clear the output screen. It is used in Turbo C/C++.

13. Difference between puts() and gets() functions?

Ans: puts() is used to display string. gets() is used to take string input.

14. Define sizeof() function with example?

Ans: sizeof() returns memory size of data type or variable. Example: sizeof(int).

CHAPTER 11 : DECISION CONSTRUCTS

1. Define control structure?

Ans: Control structure controls the flow of execution in a program. It includes decision-making and looping statements.

2. Write the limitation of if statement with example?

Ans: If statement handles only one condition at a time. Example: checking only if marks > 50.

3. Difference between if and if else statement?

Ans: If executes code when condition is true. If-else provides alternative block when condition is false.

4. Define conditional operator? With example?

Ans: It is a shorthand of if-else statement. Example: a>b ? a : b.

5. Define compound statement?

Ans: A compound statement is a group of statements inside { }. It is treated as a single unit.

6. Define delimiters in C language?

Ans: Delimiters are symbols used to separate program elements. Example: comma, semicolon, braces.

7. Define nested if with its syntax?

Ans: Nested if is an if statement inside another if. Syntax: if(condition){ if(condition){ } }.

8. Why default label is used in switch statement?

Ans: Default label is used when no case matches. It provides an alternative output.

9. Why break keyword used in switch. What happened if it is missed?

Ans: Break stops execution of switch cases. If missed, all cases below will execute.

10. Define switch statement. Also write two advantage?

Ans: Switch is a selection structure used for multiple conditions. It is faster and easier than multiple if-else.

11. Draw flowchart if else if statement?

Ans: Flowchart: Start -> Condition check -> True/False -> Multiple decisions -> End.

CHAPTER 13 : FUNCTION IN "C"

1. Define function. Also write 2 benefit?

Ans: A function is a block of code that performs a specific task. It improves code reusability and makes programs easy to manage.

2. Difference between Actual and Formal parameters?

Ans: Actual parameters are values passed in function call. Formal parameters are variables in function definition.

3. Difference between function definition and declaration with example & syntax?

Ans: Declaration tells function name and return type. Definition contains actual code of function.

4. Difference between built-in function and user defined function?

Ans: Built-in functions are predefined in libraries. User-defined functions are created by programmers.

5. What is function header with example and syntax?

Ans: Function header defines return type, name, and parameters. Example: int sum(int a, int b)

6. What is function definition? Also write components?

Ans: Function definition contains function body. Components are header, parameters, and body.

7. How does function return value?

Ans: A function returns value using return statement. It sends output back to calling function.

8. What do you mean by function call?

Ans: Function call is the process of executing a function. It passes control to function definition.

9. What is the life time of local variable and global variable?

Ans: Local variables exist only inside function. Global variables exist throughout program execution.

10. Difference between local and global variable with example?

Ans: Local variable is declared inside function. Global variable is declared outside all functions.

11. Define function argument?

Ans: Function arguments are values passed during function call. They are used by function to perform tasks.

12. Difference between formal parameter and actual parameter?

Ans: Formal parameters are in function definition. Actual parameters are values passed in function call.